
Q-OPS: Evidence-Aware Quantum–Classical Routing for Trustworthy Human–AI Staffing

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Abstract

1 Organizations allocating work among humans, AI systems, and human–AI teams
2 must balance service, cost, limited review capacity, correlated failures, and hard
3 operating requirements. Indiscriminate adoption of an uncertain computational
4 technology creates a second trust problem: a promising candidate is not sufficient
5 evidence that the technology should be invoked. Q-OPS treats quantum compu-
6 tation as an optional expert inside an application-oriented decision architecture.
7 Strong classical methods first establish feasibility and remaining headroom; an
8 evidence gate identifies a selective quantum pilot; every candidate is independently
9 checked; unsupported use falls back to the classical incumbent. In the frozen study,
10 the quantum branch audits a retained-state distribution and does not replace that in-
11 cumbent. Among 32 synthetic held-out instances, 9/32 (28.13%) meet the recorded
12 post-simulation evidence gate. Across all retained-core comparisons, the matched-
13 budget optimum-sampling probability ratio is $6.831\times$ relative to uniform feasible
14 sampling. The shared evaluator records zero disallowed-mode violations, while
15 only 27/32 classical incumbents satisfy the stricter composite-feasibility predicate.
16 Earlier C1.0–C1.1 development stages and the frozen depth comparison preserve
17 negative evidence: strong classical controls can remove apparent headroom, and
18 $p = 1$ exceeds $p = 2$ on 28/32 cases under unequal parameter-search budgets. The
19 present result is therefore statevector-level conditional algorithmic evidence, not
20 real-QPU, wall-clock, scaling, universal, or application-level quantum advantage.
21 Q-OPS illustrates a general rule for trustworthy emerging-computation adoption:
22 compare against credible alternatives, invoke selectively, certify independently, and
23 retain a safe fallback.

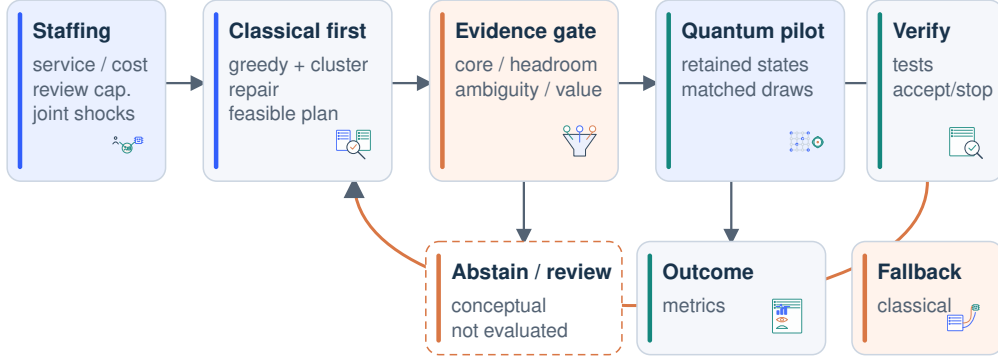
24 1 Introduction

25 Future service organizations allocate work among humans, AI systems, and human–AI teams. Each
26 choice changes expected service, cost, capacity, accountability, review burden, and exposure to
27 correlated AI failures. Human–AI staffing is therefore an economic resource-allocation problem with
28 hard operational and institutional constraints, not merely a scheduling toy. Research on human–AI
29 interaction calls for explicit authority and oversight boundaries [1, 26]; service operations supplies
30 capacity and reliability models [11]; and sociotechnical analysis cautions that efficiency or a compact
31 fairness proxy cannot stand in for stakeholder welfare [27].

32 Quantum optimization introduces a second adoption decision. Even if a statevector experiment
33 concentrates probability on good candidates, using the quantum component may be irrational once
34 classical alternatives, validation, delay, review, and risk costs are counted. Small noiseless demonstra-
35 tions also do not establish QPU performance, wall-clock advantage, or scaling [3, 5, 22]. Rigorous
36 optimization benchmarks consequently emphasize standardized instances, strong counterfactuals,

Trustworthy application-level architecture

Solid paths are implemented in the frozen audit; dashed actions are architectural extensions.



Evidence boundary: quantum sampling can inform a candidate distribution; live labels, gates, certification, and fallback

Figure 1: **Trustworthy application-level architecture.** Strong classical alternatives precede the recorded evidence gate; the quantum branch is optional; a shared classical evaluator and fallback bound operational acceptance. Dashed abstention/review is conceptual and was not evaluated. Clip art is semantic only; labels, arrows, gates, and evidence states are editable vector layers.

37 solution quality, resources, and reproducibility [17]; application-centric frameworks additionally ask
38 whether the complete workflow improves a meaningful task [9].

39 Q-OPS moves the question one level upward: *when is there enough evidence to invoke quantum*
40 *computation at all?* The architecture in Figure 1 treats quantum computation as an optional expert
41 after a strong classical stage and before an independent certificate and fallback. It distinguishes system-
42 level benchmarking (a fixed stack), algorithm-level benchmarking (controlled solver comparison),
43 and application-level benchmarking (incremental decision value for the same organizational problem).
44 The frozen study reaches the second level only on selectively retained cores inside an application-
45 oriented architecture.

46 The paper is organized around exactly three research-question–contribution pairs:

47 **RQ1. How can Human–AI staffing jointly represent service, cost, review scarcity, collabora-**
48 **tion, correlated failures, and hard requirements?** We contribute a synthetic, governance-
49 constrained benchmark with human-only, AI-only, and collaborative modes, empirical chance
50 constraints, disruption scenarios, and an explicit composite feasibility predicate.

51 **RQ2. When should a trustworthy decision system remain classical or run a quantum pilot?** We
52 contribute an evidence-aware routing architecture driven by a multidimensional instance state,
53 remaining classical headroom, ambiguity, feasibility, certification, and fallback. Abstention
54 and human review are architectural actions, not evaluated policies in the frozen experiment.

55 **RQ3. What conditional value survives strong counterfactuals and application-aware evidence**
56 **boundaries?** We contribute a frozen, null-inclusive evaluation with matched sampling budgets,
57 negative development stages, exact shared-field reproduction, and an evidence ladder that
58 separates sampling improvement from optimization, application, and welfare claims.

59 The headline result is deliberately two-sided. Nine of 32 held-out instances (28.13%) satisfy the
60 recorded post-simulation gate; across all 32 retained-core comparisons, $p = 2$ statevector sampling
61 has a $6.831 \times$ ratio of mean optimum probability over uniform feasible sampling. Independent
62 evaluation records zero disallowed-mode violations, but composite feasibility holds for only 27/32
63 classical incumbents. The gate uses an observed quantum statistic and is not a prospective cost-saving
64 router; $p = 1$ exceeds $p = 2$ on 28/32 cases under unequal search budgets; and stronger full-problem
65 classical or hardware comparisons remain absent. A trustworthy system should maximize *justified*,
66 not maximal, quantum use.

Table 1: Benchmark and decision environment. All coefficients are frozen synthetic choices, not field-calibrated welfare weights.

Family	Benchmark encoding	Interpretation
Decision	Human-only, AI-only, or human–AI collaboration	Accountability and automation choice
Reliability	24 correlated disruption scenarios; empirical chance constraints	Customer continuity under shocks
Oversight	Finite review capacity; collaboration uses partial review	Reviewer workload and escalation
Workflow	Cluster complementarity, concentration, and mode friction	Organizational coordination
Objective	Expected service value minus cost, overflow, and penalties	Transparent trade-offs, not welfare
Acceptance	Composite feasibility, shared evaluator, classical fallback	Auditable operational boundary

2 Related Work and Human–AI Staffing Formulation

Intersecting literatures. Algorithm selection asks which solver fits an instance [18, 23]; selective prediction and learning to defer add abstention or handoff under uncertainty [12, 19]. QAOA and alternating operators provide parameterized distributions and constraint-aware mixers [8, 13], while warm starts inject classical information [7]. QOBLIB argues for model-independent instances, strong classical records, standardized metrics, and transparent resource reporting [17]. Q-OPS connects these components to a staffing decision in which certification and fallback remain classical. Its integration claim is qualified: it does not introduce a new circuit family, prove a welfare theorem, or establish field validity.

Staffing benchmark. For tasks $i \in I$, choose a legal mode $y_i \in D_i \subseteq \{H, A, C\}$ for human-only, AI-only, or collaboration. Tasks have duration d_i , priority p_i , value v_i , region $r(i)$, and mode-dependent cost c_{i,y_i} . Review-required AI work consumes one unit and collaboration one quarter,

$$C(y) = \sum_i d_i c_{i,y_i}, \quad R(y) = \sum_i r_i (\mathbb{I}[y_i = A] + .25\mathbb{I}[y_i = C]), \quad (1)$$

with review overflow $R_+(y) = [R(y) - B_R]_+$. Workflow clusters receive a thresholded collaboration benefit, an AI-only concentration penalty, and a mode-friction penalty; the complete frozen formula appears in Appendix A.

For each correlated disruption scenario $\omega \in \Omega$, a shared evaluator computes weighted service and regional capacity overflow. Empirical service and capacity failure rates are

$$q_s(y) = |\Omega|^{-1} \sum_{\omega} \mathbb{I}[\text{service}_{\omega}(y) < .70], \quad q_c(y) = |\Omega|^{-1} \sum_{\omega} \mathbb{I}[O_{\omega}(y) > 0]. \quad (2)$$

Composite feasibility is $F(y) = \mathbb{I}[H(y) = 0, R_+(y) = 0, q_s(y) \leq .20, q_c(y) \leq .20]$, where H counts disallowed modes. The score combines expected value minus cost and overflow, workflow complementarity, a regional AI-exposure regularizer, review overflow, and chance-constraint penalties. All coefficients are frozen synthetic design choices, not learned welfare weights, workplace standards, or fairness guarantees.

The benchmark makes stakeholder consequences visible without claiming them as measured benefits. Customers depend on service continuity; workers face review load and inappropriate automation risk; organizations face cost, accountability, and resilience; society faces the broader question of disciplined technology adoption. Lower cost, more AI assignments, or more collaboration need not imply higher welfare.

3 Evidence-Aware Quantum–Classical Routing

Observable decision state. Let

$$z = [n, d, \rho, g, a, r, s, c] \quad (3)$$

summarize dimension, constraint density, interaction structure, remaining classical headroom, disagreement among strong classical methods, review scarcity, disruption severity, and computational cost. These components form an instance-difficulty vector rather than treating qubit count as difficulty. The frozen implementation observes only a subset directly; the complete vector is the intended application-level interface and its unevaluated components are future work.

Classical counterfactual and retained core. A legal coordinate-greedy solution is challenged by cluster destroy/repair with exact neighborhood enumeration. Let y^G, y^C be the incumbents, $\Delta = S(y^C) - S(y^G)$, d_G their normalized disagreement, and v_R restart-score variance. The frozen ambiguity score is

$$a = \min \left\{ 1, 2.8d_G + 5 \left[\frac{\Delta}{\max(|S(y^C)|, 1)} \right]_+ + 4 \frac{\sqrt{v_R}}{\max(|S(y^C)|, 1)} \right\}. \quad (4)$$

Four interacting tasks are retained and every legal assignment is classically evaluated with the incumbent fixed outside the core. Strictly feasible states form $\mathcal{Z}$; an empty set triggers an explicitly labeled least-violation fallback and never becomes “feasible.” This exact retained-core enumeration is both a baseline and a certificate.

Optional quantum pilot. The warm state is the retained assignment nearest y^C . A feasible-subspace statevector QAOA distribution uses the retained-state adjacency as mixer and standardized score as cost. Uniform feasible sampling and QAOA receive the same 128-draw sampling budget. The empirical frozen gate requires a strict core, $a \geq .08$, classical headroom/disagreement, and $p_2^* \geq \max(.18, 1.4p_U^*)$. Since it consumes p_2^* , it is a *post-simulation evidence gate*, not a deployable pre-execution policy.

The frozen quantum branch audits the distribution and does not replace the cluster-repair incumbent. The prospective policy is instead

$$\pi(z) = \begin{cases} \text{Quantum Pilot,} & F(z) = 1, H(z) > \tau_H, A(z) > \tau_A, \hat{V}_Q(z) > \tau_Q, \\ \text{Classical,} & \text{strong classical evidence is sufficient,} \\ \text{Abstain / Human Review,} & \text{evidence is insufficient or risk is unacceptable,} \end{cases} \quad (5)$$

where abstention/review are conceptual extensions. Economically, quantum invocation is justified only when

$$\mathbb{E}[\Delta W \mid z] > C_Q + C_{\text{validation}} + C_{\text{risk}} + C_{\text{review}}. \quad (6)$$

Q-OPS estimates none of these welfare terms; the inequality is a decision scaffold, not a theorem.

Certificate and evidence ladder. Every candidate passes the same classical scenario evaluator before acceptance; otherwise the system returns the classical incumbent or escalates. We distinguish five evidence levels: L1 quantum feasibility; L2 conditional algorithmic value under controlled budgets; L3 hardware-level value; L4 end-to-end wall-clock value; and L5 application-level economic value. The present paper reaches L2 across frozen retained-core comparisons only; 9/32 cases satisfy the post-simulation evidence gate, but no prospective adoption policy was evaluated.

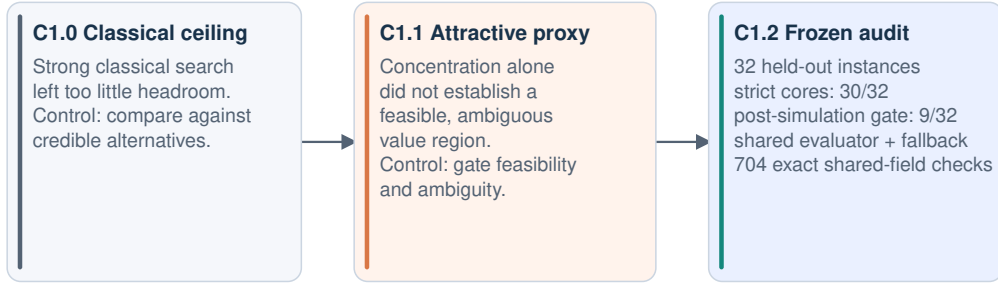
4 Experimental Design

The immutable C1.2 matrix crosses task counts $\{12, 18\}$, four synthetic stress levels, and held-out seeds $\{101, 211, 307, 401\}$ for 32 instances. Development seeds 11 and 23 are excluded. Each instance uses 24 correlated scenarios, three regions, three-task workflow clusters, service target .70, chance tolerance .20, review capacity, and a four-task retained core. QAOA depths $p = 1, 2$ are frozen; $p = 1$ receives a $13 \times 11 = 143$ -point grid and $p = 2$ receives 30 seeded random trials, so depth comparisons are descriptive rather than an equal-optimizer-budget ablation.

Uniform and $p = 2$ retained-core probabilities each receive 128 draws. Prespecified outcomes include strict-core availability, classical gain and ambiguity, optimum mass, first-hit draw (miss coded 129), near-optimal coverage, gate status, composite feasibility, and hard violations. Percentile 95% bootstrap intervals resample paired instances 10,000 times. Sampling probability, best-of-budget assignment, practical runtime, application outcomes, and welfare remain distinct estimands.

Evidence evolution: controls added because earlier signals were insufficient

C1.0--C1.1 are qualitative development evidence; only C1.2 carries frozen held-out results.



Scientific argument Apparent value can vanish without feasibility, strong classical controls, and relevance.

Remaining boundary Routing, QPU/noise, end-to-end cost, and stakeholder outcomes remain open.

Figure 2: **Evidence evolution rather than a decorative timeline.** C1.0 and C1.1 identify why strong classical controls, strict feasibility, ambiguity, matched comparison, certification, and fallback are necessary. Only the immutable C1.2 held-out evaluation supplies headline numbers.

139 The development sequence is preserved rather than rewritten as a success-only narrative. C1.0
140 left too little useful headroom after strong classical search. C1.1 showed probability concentration
141 without jointly establishing strict feasibility and classical ambiguity. C1.2 froze feasibility, head-
142 room/ambiguity, matched retained-state comparison, selective gating, independent certification, and
143 fallback before the held-out run. Figure 2 shows why each control is necessary; only C1.2 carries
144 headline numbers.

145 The canonical configuration SHA-256 is 7f07459abc9e...7715e53c; the identifying source
146 commit is recorded only in the double-blind external audit. A fresh runner produced 32 rows.
147 Across the 22 shared fields and 32 records, all 704 comparisons match the archived table exactly
148 (maximum absolute error 0). Eight archived diagnostics are no longer emitted, so the claim is exact
149 *shared-field* reproduction, not byte-identical CSV reproduction. New router baselines, gate ablations,
150 risk-coverage curves, QPU runs, and stronger full-problem solvers were not reproducible from the
151 frozen protocol without changing the study; they are therefore preregistered evaluation priorities
152 rather than fabricated results.

153 5 Results and Analysis

154 **Classical structure and feasibility.** Strict feasible retained states exist on 30/32 instances. Cluster
155 repair improves over coordinate greedy on 25/32, with mean score gain 17.019 (paired bootstrap 95%
156 CI [12.524,21.552]); collaboration rises descriptively from .475 to .664. These findings establish
157 a meaningful classical counterfactual inside the frozen benchmark, but not superiority over mixed-
158 integer programming, CP-SAT, exhaustive full-instance optimization, or a broad ALNS portfolio.
159 Only 27/32 cluster incumbents satisfy composite feasibility even though all 32 record zero disallowed-
160 mode violations, underscoring why certification and fallback cannot be collapsed into a single metric.

161 **Selective conditional evidence.** Nine of 32 cases (28.13%) pass the recorded gate. Across all 32
162 retained-core comparisons, mean optimum mass is .0223 for uniform and .1522 for $p = 2$, a ratio of
163 $6.831 \times$ and paired difference .1299 (95% CI [.0991,.1640]); the first-hit proxy changes from 36.50
164 to 12.94 draws (paired difference -23.56 [$-39.28, -8.69$]). The near-optimal coverage difference is
165 $+0.0443$ [$-.0208, .1120$], whose interval crosses zero. Figure 3 maps all observations: low ambiguity
166 is classically sufficient; high ambiguity without adequate concentration is evidence-insufficient; the
167 upper-right region contains quantum candidates subject to the remaining gates.

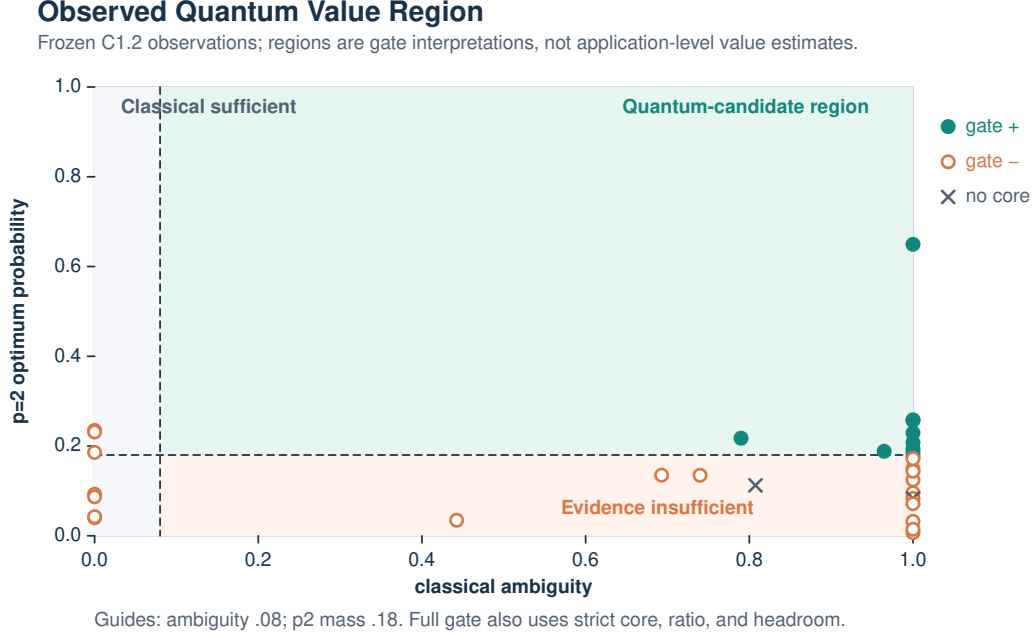


Figure 3: **Observed Quantum Value Region.** Each frozen instance is plotted by classical ambiguity and $p = 2$ optimum probability. Filled points pass the complete post-simulation gate; open points fail it; crosses lack a strict core. Shaded regions interpret two gate dimensions only and do not encode economic value or a prospective routing policy.

Table 2: Frozen C1.2 benchmark results. Intervals are 95% paired bootstrap intervals where available; “–” means no interval was specified.

Metric	Estimate	95% CI / denominator	Scope
Held-out matrix	32	–	Frozen; all cases retained
Strict feasible core	30/32	–	Two least-violation fallbacks
Cluster repair gain	17.019	[12.524, 21.552]	Paired instance bootstrap
Uniform optimum mass	0.0223	[0.0147, 0.0350]	All retained cores
$p = 2$ optimum mass	0.1522	[0.1161, 0.1940]	Noiseless statevector
$p = 2$ minus uniform	0.1299	[0.0991, 0.1640]	Matched retained state set
Evidence-gate positive	9/32 (28.13%)	[12.50%, 43.75%]	Post-simulation; not prospective
Composite-feasible incumbent	27/32	–	Stricter than mode legality
Disallowed-mode violations	0/32	–	Not total safety or feasibility

Negative and null evidence. The frozen ordered gate failures are two cases without a strict core, seven with low ambiguity, and fourteen with inadequate $p = 2$ concentration. The low gate-positive rate is a trust feature only in the limited sense that the recorded gate rejects universal use; because the gate is post-simulation, it does not reduce quantum calls. Mean $p = 1$ optimum mass is .2492 and $p = 1 > p = 2$ on 28/32 cases, but the parameter-search budgets differ, so neither deeper-is-better nor shallower-is-better is identified. Uniform feasible sampling is a controlled baseline, not a strong simulated-annealing, tabu, beam-search, or distributional classical portfolio.

What routing evaluation still requires. A leakage-free test must freeze a router using pre-execution features and compare Always Classical, Always Quantum, Random, Evidence-Aware, and a hindsight oracle on held-out instances. Coverage $N^{-1} \sum_i \mathbb{I}[\pi_\tau(z_i) = Q]$, conditional gain, compute/review cost, violations, certification, and fallback should be reported as thresholds vary. The current data cannot support that curve without retrofitting a prospective policy, so Q-OPS reports the test design and limitation rather than inventing points.

6 Trustworthiness, Limitations, and Conclusion

Trust mechanisms and human implications. The architecture exposes the classical counterfactual, records why a quantum branch is considered, verifies every candidate with an independent evaluator, and preserves fallback. These mechanisms support auditability; they do not certify safe deployment. Customers need continuity under shocks, workers need limits on automation and review overload, and managers need accountable cost–service trade-offs. Society benefits only if adoption standards prevent performance proxies from displacing rights, welfare, and institutional legitimacy. Lower staffing cost is not automatically greater welfare; more AI or collaboration is not automatically desirable.

Scientific limits. The benchmark is synthetic: costs, success rates, disruptions, and penalties are not estimated from organizations or affected workers. There are no human subjects or personal data. Regional AI exposure is not fairness, collaboration is not worker benefit, and zero hard violations is not workplace safety. Quantum evidence is limited to four retained ternary decisions, exact enumeration, noiseless statevectors, unequal depth-search budgets, no circuit decomposition, no QPU, no noise, no wall-clock or energy accounting, and no scaling test. Stronger classical full-problem and retained-core samplers could reduce or eliminate the observed signal. Bootstrap intervals describe these 32 frozen instances, not a population of workplaces.

Application-level path. Progress from L2 to L5 requires a prospective router, equal and widened classical budgets, hardware/noise studies, end-to-end time/cost accounting, operational outcomes, stakeholder-defined measures, and field-valid data with governance. Any future deployment needs human override, privacy controls, distribution-shift monitoring, and a documented accountability owner. NIST and ISO frameworks are governance references, not certifications [15, 16, 21].

Q-OPS consequently supports a narrow but useful conclusion: a frozen statevector experiment identifies conditional sampling value across retained-core comparisons, while a post-simulation gate marks a selective evidence region and the evaluator records zero disallowed-mode violations. It does not establish composite feasibility in every case or quantum advantage at the system, application, economic, or welfare level. The value of an emerging computational component depends not merely on whether it can generate promising candidates, but on whether a decision system can identify when its incremental value justifies adoption, compare it against credible alternatives, certify its outputs, and preserve a safe fallback when evidence is insufficient.

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A Benchmark formulas, meanings, and precedents

This appendix distinguishes four evidence statuses: *implemented* means the public evaluator computes the quantity; *synthetic* means its coefficient is generated or frozen rather than estimated from field data; *interpretive* means we supply a real-world analogy; and *validated* is deliberately not asserted for organizational welfare, safety, or fairness.

For scenario ω , let D_ω be demand, $h_{r\omega}$ and $a_{r\omega}$ regional human/AI factors, and u_ω the common AI shock. With clipped success to $[0, 1]$,

$$s_{i\omega}(H) = \bar{s}_i^H h_{r(i)\omega}, \quad (\ell^H, \ell^A)(H) = (1, 0), \quad (7)$$

$$s_{i\omega}(A) = \bar{s}_i^A a_{r(i)\omega} u_\omega, \quad (\ell^H, \ell^A)(A) = (0, 1), \quad (8)$$

$$s_{i\omega}(C) = \bar{s}_i^C (.55 h_{r(i)\omega} + .45 a_{r(i)\omega}) (.985 + .015 u_\omega), \quad (\ell^H, \ell^A)(C) = (.45, .65). \quad (9)$$

Regional overflow is the sum of positive excesses over scenario-scaled human and AI capacities:

$$O_\omega(y) = \sum_r [L_{r\omega}^H(y) - B_r^H h_{r\omega}]_+ + [L_{r\omega}^A(y) - B_r^A a_{r\omega}]_+, \quad (10)$$

where $L_{r\omega}^x = \sum_{i:r(i)=r} d_i D_\omega \ell^x(y_i)$. These equations correct a tempting but inaccurate simplification: the service constraint is aggregate and scenario-based, not a separate chance constraint for every task.

Table 3: Complete benchmark formula glossary. Every row is implemented in the frozen evaluator.

Factor	Implemented definition	Plain-language / real-world role	Evidence and literature
Mode	$y_i \in \{H, A, C\}; A \notin D_i$ for human-mandatory i	Each task is human-only, AI-only, or collaborative; some retain human responsibility.	Directly enumerated domain; human–AI interaction design motivates explicit authority boundaries [1, 26].
Hard violations	$H(y) = \sum_i \mathbf{1}[y_i \notin D_i]$	Counts disallowed assignments; this is narrower than composite feasibility.	Evaluator and final certificate; governance context [16, 21].
Base cost	$C(y) = \sum_i d_i c_{i, y_i}$	Duration-weighted cost of the chosen operating mode.	Scenario-independent term in the evaluator; resource allocation tradition [11].
Review load	$R(y) = \sum_i r_i (\mathbf{1}[y_i = A] + .25 \mathbf{1}[y_i = C])$	AI-only work consumes a full review unit; collaboration consumes one quarter.	Synthetic coefficient, not empirically calibrated; makes oversight capacity explicit [21].
Review overflow	$R_+(y) = [R(y) - B_R]_+$	Review demand above the available audit budget.	Penalized by 9 in the score; operational bottleneck construct.
Cluster reward	$\sum_k B_k \mathbf{1}[n_k^C \geq h_k]$	Collaboration creates value only after a workflow-level threshold.	Implemented clusters of at most three tasks; complementarity motivation [10, 26].
All-AI penalty	$\sum_k A_k \mathbf{1}[\forall i \in k : y_i = A]$	A fully automated workflow incurs a governance/robustness cost.	Synthetic penalty, increasing with stress level; not a causal harm estimate.

Factor	Implemented definition	Plain-language / real-world role	Evidence and literature
Handoff penalty	$\sum_k J_k(\{y_i : i \in k\} - 1)_+$	More modes inside one cluster create coordination overhead.	Implemented workflow mixing cost; organizational interdependence motivation [14, 28].
AI exposure dispersion	$P_{\text{exp}}(y) = 2 \text{sd}_r(a_r), a_r = I_r ^{-1} \sum_{i \in I_r} \mathbf{1}[y_i = A]$	Penalizes uneven AI-only rates across modeled regions.	Coded diagnostic, <i>not</i> validated fairness; sociotechnical cautions [2, 20, 27].
Scenario loads	$L_{r\omega}^H = \sum_{i \in r} d_i D_\omega \ell^H(y_i),$ $L_{r\omega}^A = \sum_{i \in r} d_i D_\omega \ell^A(y_i),$ with $(\ell^H, \ell^A) = (1, 0), (0, 1), (.45, .65)$	Human and AI capacity draw changes by mode and correlated demand scenario.	Exact frozen coefficients; chance-constrained operations motivation [6, 11].
Scenario success	$s_{i\omega} = \bar{s}_{i, y_i}$ times regional human/AI factors and AI shock; collaboration blends factors .55/.45	Mode success changes with regional and common disruptions.	Exact generator/evaluator equations in Appendix A; synthetic, not field-estimated.
Service risk	$q_s(y) = \Omega ^{-1} \sum_\omega \mathbf{1}[\rho_\omega(y) < .70],$ with $\rho_\omega = \sum_i p_i s_{i\omega} / \sum_i p_i$	Fraction of scenarios whose priority-weighted service falls below target.	Target .70, tolerance $\epsilon = .20$; empirical chance constraint [6].
Capacity risk	$q_c(y) = \Omega ^{-1} \sum_\omega \mathbf{1}[O_\omega(y) > 0]$	Fraction of scenarios with any regional human/AI capacity overflow.	Tolerance $\epsilon = .20$; overflow also priced inside scenario value.
Scenario value	$V_\omega(y) = \sum_i v_i s_{i\omega} - C(y) - 4O_\omega(y)$	Expected delivered value net of cost and scenario overflow.	Averaged over 24 scenarios.
Composite score	$S(y) = \bar{V} + \text{cluster reward} - \text{cluster penalties} - P_{\text{exp}} - 9R_+ - 600H - 150[q_s - .2]_+ - 45[q_c - .2]_+$	A single score ranks assignments while retaining explicit diagnostics.	Exact frozen implementation; coefficients are benchmark parameters, not learned welfare weights.
Composite feasible	$F(y) = \mathbf{1}[H = 0, R_+ = 0, q_s \leq .2, q_c \leq .2]$	Requires mode legality, review capacity, service reliability and capacity reliability.	Used for strict retained states and reported separately from hard-mode legality.
Collaboration rate	$n^{-1} \sum_i \mathbf{1}[y_i = C]$	Describes how often the collaborative mode is selected.	Descriptive outcome only; it is not itself a success metric.

305 **Generator distributions.** The generator assigns three-task clusters, up to three modeled regions,
306 task values in [9, 18], human success in [.94, .99], AI success based on [.84, .955] with a sensitivity
307 decrement, and collaborative success above the better base mode by a sampled margin, capped at
308 .997. Costs, capacities, review requirements, cluster coefficients and disruption strength are seeded.
309 The four stress levels increase sensitive-task prevalence, outage probability and selected penalties.
310 These ranges define a controlled stress test, not an empirical model of any named industry.

311 B Complete U–W–C–Q formula glossary

312 The mnemonic is not execution order. The classical layer encloses the experiment: it produces
313 the incumbent and retained set, while the same evaluator verifies the final assignment. The warm
314 layer feeds Q, and U/Q form a paired distributional comparison. Figure 1 visualizes the enclosing
315 application architecture; Table 4 records every implemented strategy and metric.

Table 4: Complete U–W–C–Q strategy and metric glossary.

Object	Formula / implementation	Intuitive definition	Use and precedent
Individual seed	$\arg \max_{m \in D_i} (v_i \bar{s}_{im} - d_i c_{im})$ per task	Myopic value-minus-cost initialization.	Starting point for coordinate descent.
Greedy control	Repeated best legal one-task move until no S improvement	Stronger than an all-human or random straw baseline.	Classical control; local-search precedent.
Cluster repair	Exact enumeration of one three-task cluster, sometimes plus one external task, for 18 iterations and 2 restarts	Destroy/repair coordinated assignments inside a small neighborhood.	Inspired by ALNS [24]; the implementation is intentionally reported as <i>cluster repair</i> , not a comprehensive ALNS suite.
Classical gain	$\Delta = S(y^C) - S(y^G)$	Improvement of cluster repair over greedy.	One ambiguity signal; mean and win rate are frozen outcomes.
Disagreement	$d = n^{-1} \sum_i \mathbf{1}[y_i^G \neq y_i^C]$	How differently the two classical procedures allocate tasks.	A second ambiguity signal.
Ambiguity	$a = \min\{1, 2.8d + 5[\Delta / \max(S_C , 1)]_+ + 4\sqrt{v_R} / \max(S_C , 1)\}$	Normalized evidence that the classical landscape is contested.	Threshold .08; diagnostic, not calibrated probability. Algorithm selection context [18, 23].
Core score	$B_k + A_k + J_k \max(1, k - 1) + 5\bar{d}_k$	Ranks clusters by complementarity, penalties and disagreement.	Selects four task indices; not a learned selector.
Retained set	Enumerate $\prod_{i \in I} D_i$; keep $F(y) = 1$	Exact small-core assignments that remain feasible with the incumbent outside the core.	At most $3^4 = 81$ states.
Fallback set	If no strict state, retain lowest 20% of $20H + 2R_+ + q_s + q_c$	Keeps a least-violating diagnostic set when strict feasibility fails.	Used on 2/32 instances; never relabeled feasible.

Object	Formula / implementation	Intuitive definition	Use and precedent
Retained graph	$A_{zz'} = 1$ for one-mode Hamming neighbors; connect isolated nodes to nearest retained state	Mixer moves only among retained states.	Feasible alternating-operator motivation [13, 25].
Warm state	$z_W = \arg \min_{z \in \mathcal{Z}} d_H(z, y_I^C); \psi_0(z) \propto e^{-.35 d_H(z, z_W)}$	Starts near the cluster-repair incumbent while retaining support elsewhere.	Warm-start motivation [7].
Hamiltonians	$H_C = \text{diag}((E - \bar{E})/\sigma_E), E = -S; H_M = -A$	Phase by standardized cost; mix along retained edges.	Statevector implementation; no gate compilation or noise model.
QAOA state	$ \psi_p\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell H_M} e^{-i\gamma_\ell H_C} \psi_0\rangle$	Alternates cost and mixer evolution.	QAOA/alternating-operator line [4, 8, 13].
Parameter search	$p = 1$: 13×11 grid; $p = 2$: 30 seeded random trials; lexicographic objective $(\mathbb{E}[E], -M_{.05}, -p^*)$	Frozen but unequal searches at the two depths.	Supports reporting each depth, not a causal depth comparison.
Uniform baseline	$P_U(z) = 1/ \mathcal{Z} $	Uninformed sampling on the same retained set.	Isolates concentration from feasibility filtering.
Optimum mass	$p^* = \sum_{z: E(z)=E_{\min}} P(z)$	Probability of drawing any exactly optimal retained state.	Primary statevector concentration metric.
Near-optimal mass	$M_{.05} = \sum_{z \in \mathcal{N}_{.05}} P(z), \mathcal{N}_{.05} = \{E \leq E_{\min} + .05 \text{ range}(E)\}$	Probability inside a 5%-of-range energy band.	Secondary optimizer tie-break and descriptive metric.
First hit	$T = \min\{b : z_b \text{ optimal}\}; T = 129$ on a miss, budget 128	Draw index of the first optimum under one seeded matched sample.	Sample-efficiency proxy, not wall-clock runtime.
Coverage	distinct sampled near-optimal states / all near-optimal retained states	Diversity inside the near-optimal band.	Paired difference interval crosses zero.
Evidence gate	$F_I \wedge a \geq .08 \wedge p_2^* \geq \max(.18, 1.4p_U^*) \wedge (\Delta > 0 \vee d \geq .05)$	Flags frozen instances with strict feasibility, ambiguity and sufficient simulated concentration.	Computed <i>after</i> QAOA; not an invocation policy.
Certificate	Re-evaluate the accepted full assignment with the shared scenario evaluator	Separates quantum evidence from operational acceptance.	In the current experiment the cluster-repair incumbent remains the final/fallback assignment.

Parameter-selection objective. For every frozen parameter candidate, statevector probabilities are ranked lexicographically by expected energy, then negative near-optimal mass, then negative exact-optimum mass. Thus p^* is not the sole optimization objective. The $p = 1$ grid and $p = 2$ random search also have unequal cardinality. Reporting the strong $p = 1$ result is therefore part of the audit rather than an ablation.

No production quantum assignment. The implementation computes an exact best retained-core state for reference and distributional sampling metrics, but it does not replace the cluster-repair incumbent with a sampled quantum candidate. “Classical certificate” in this paper means that any accepted full assignment is evaluated by the common scenario evaluator; in the frozen run, the operational incumbent remains classical. This distinction prevents probability concentration from being misreported as an implemented end-to-end policy improvement.

C Frozen protocol and pseudocode

Locked runner. For each tuple $(n, \text{stress}, \text{seed})$ in the frozen matrix:

1. generate the synthetic tasks, clusters, capacities and 24 correlated scenarios;
2. run coordinate greedy and cluster repair, preserving scores, assignments and restart variance;
3. rank clusters, select four task indices, exactly enumerate their legal assignments, and retain composite-feasible states (or explicitly marked least-violating fallback states);
4. compute the uniform distribution, the warm state, and optimized noiseless QAOA distributions at $p = 1, 2$;
5. draw 128 samples from uniform and $p = 2$ with frozen method-specific seeds;
6. calculate ambiguity and the post-simulation evidence gate; and
7. serialize every instance, including null and gate-negative cases, before aggregate bootstrap inference.

Inference. The instance is the paired unit. For a contrast d_j over $j = 1, \dots, 32$, the reported percentile interval is the 2.5th and 97.5th percentiles of 10,000 means formed by resampling instance indices with replacement. Seeds 7001–7004 are fixed for gain, optimum-mass difference, first-hit

Table 5: Frozen experimental protocol.

Component	Frozen value
Factorial matrix	$n \in \{12, 18\}$; stress/doping $\in \{0, 1, 2, 3\}$; seeds $\{101, 211, 307, 401\}$; 32 instances total
Scenarios	24 correlated scenarios per instance; service target .70; tolerance .20; capacity scale .92
Classical	Coordinate greedy; cluster destroy/repair, 18 iterations, 2 restarts
Retained core	4 variables; ternary legal domains; exact retained-core enumeration; at most 81 states
Quantum	Noiseless feasible-subspace statevector; $p = 1, 2$; $p = 1$ grid 143 points; $p = 2$ seeded random search 30 points
Sampling	128 replacement draws per U/Q distribution; different frozen seeds; miss coded 129
Inference	Instance is the paired unit; percentile paired bootstrap; 10,000 resamples; all prespecified nulls retained
Config digest	7f07459abc9e686934be1ee022768eb82 a9687137cc86d4d7ea0b7727715e53c
Source snapshot	Anonymous frozen implementation artifact; identifying commit retained in the external audit only

342 difference and coverage difference. These intervals quantify the frozen matrix, not a random sample
343 of real organizations.

344 **Compute.** The independent verification used Python 3.12.13 and NumPy 2.3.5 on Linux 6.18.35,
345 an AMD EPYC 9V74 allocation with 9 logical CPUs and 21 GiB RAM. The complete 32-instance
346 locked rerun took 27.64 seconds wall-clock and no GPU. The archived manifest records Python 3.13.5
347 and NumPy 2.3.5 but not original CPU or wall time; this missing historical metadata is disclosed
348 rather than reconstructed.

349 D Additional results and diagnostic partitions

Table 6: Ordered post-simulation evidence-gate partition.

First failing/succeeding condition	Count	Interpretation
No strict feasible retained state	2	Violation-ranked fallback; cannot support strict-core evidence.
Classical ambiguity below .08	7	Strong classical methods give little reason to focus on disagreement.
Concentration below frozen threshold	14	QAOA $p = 2$ does not meet both absolute and relative mass thresholds.
All evidence conditions pass	9	Post-simulation evidence-positive subset; still not a deployment decision.

350 **Depth and matched draws.** Across instances, mean optimum masses are .022278 (uniform),
351 .249182 ($p = 1$), and .152190 ($p = 2$). The $p = 2$ value exceeds uniform on 30/32; $p = 1$ exceeds
352 $p = 2$ on 28/32. Since the parameter searches are unequal and selected by expected energy before
353 optimum mass, this is a warning against interpreting depth as the sole causal factor.

354 Uniform and $p = 2$ near-optimal coverage means are .919271 and .963542. Their paired mean
355 difference .044271 has 95% interval $[-.020833, .111979]$. We therefore report the direction but
356 make no positive coverage claim. The first-hit means of 36.50 and 12.9375 are sensitive to the frozen
357 draw stream and miss code; they complement, rather than replace, the exact probability-mass result.

358 **Feasibility decomposition.** Thirty instances have at least one strict feasible retained state; two use
359 fallback states. All 32 cluster-repair incumbents have zero explicit disallowed-mode violations, while
360 27 satisfy the stricter conjunction of review capacity and empirical service/capacity tolerances. Any
361 abstract phrase “zero final hard violations” should be read as zero mode-domain violations only.

Gate interpretation. The ordered partition in Table 6 prevents double counting. Its 9 positive instances meet all frozen predicates after QAOA. A real invocation router would have to replace p_2^* with pre-execution features and be evaluated on new held-out instances. Until then, “gate rate” is an evidence summary, not a rate of saved QPU calls.

E Reproducibility and claim traceability

The supplementary source contains the frozen configuration, archived 32-row evidence table, aggregate summary, rerun summary, configuration digest, figure generator, and a shared-field comparison script. For anonymous review, the public location is represented as the anonymous supplementary artifact; the camera-ready source should replace this placeholder with the permanent repository URL and commit.

Minimal reproduction. From the public source snapshot, install NumPy and pytest, set the repository root on PYTHONPATH, and run the three targeted test modules for the C1.1/C1.2 generator, evaluator and locked protocol. Then execute the locked runner with its explicit confirmation flag and a new empty output directory. Finally run `scripts/audit_reproduction.py` from this package against the archived and fresh CSVs. The independent audit obtained 32/32 rows, 704/704 exact shared-field matches and maximum absolute error zero.

The archived CSV has 30 columns; the current public runner emits 22. The eight archive-only diagnostics are restart variance, composite incumbent feasibility, retained-state count, fallback use, $p = 2$ near-optimal mass, effective support, and uniform/QAOA sample diversity. Because the schemas differ, the package does not claim byte identity. The richer archive remains the source for composite-feasibility and depth diagnostics; every field currently emitted by the runner is exactly reproduced.

Table 7: Claim-to-artifact traceability.

Claim / object	Public artifact or function	Verification route
Generator and evaluator	<code>qops/c12_generator.py</code> ; <code>generate_instance</code> ; <code>evaluate</code>	Exact formulas summarized in Appendix A; targeted tests cover determinism and legality.
Classical controls	<code>qops/c12_classical.py</code> ; <code>coordinate greedy</code> ; <code>cluster repair</code>	Rerun all 32 frozen instances; compare gain, disagreement and feasibility fields.
Core and statevector	<code>qops/c12_quantum.py</code> ; <code>core extraction</code> ; retained-state QAOA	Recompute retained states, $p = 1/p = 2$ masses and matched draws.
Frozen matrix and inference	<code>qops/c12_locked.py</code> ; <code>locked config JSON</code>	Explicit confirmation flag; configuration SHA-256; deterministic seeds.
Headline aggregates	Archived records CSV and summary JSON	32 archived rows; fresh run matches 704/704 values shared with the current runner exactly.
Figures and tables	<code>scripts/generate_release_assets.py</code> and copied frozen CSV	Regenerate live-text SVG masters and table inputs; export PDF; run structural and numerical audits.

Repository-wide health caveat. The targeted frozen experiment passes. A broad legacy test discovery also surfaced an unrelated sparse-tracking certificate failure, and a clean environment initially lacked pytest for a protocol module. These do not alter the C1.2 values, but they mean the snapshot should not be described as having an entirely green repository-wide test suite. The independent report records this distinction.

F Expanded limitations, ethics, and release guidance

Construct validity. The benchmark combines factors that are individually plausible but jointly synthetic. Penalty weights trade unlike quantities inside one score; they are not moral exchange rates.

392 Regional AI exposure is observed at the region level and omits protected attributes, within-region
393 heterogeneity, job quality, worker consent and downstream customer outcomes. The framework
394 should therefore be used to test algorithms, not to rank actual employees or justify automation.

395 **Quantum validity.** Exact enumeration of at most 81 retained states makes ground truth available
396 and is a virtue for auditing, but also removes the scale at which computational advantage could arise.
397 Statevector probability ignores hardware sampling noise, connectivity, compilation, calibration drift,
398 queue time and energy. Warm initialization and a retained graph are simulated as dense linear algebra;
399 their circuit cost is not estimated. A hardware study must report qubits, gates, depth, shots, error
400 mitigation, compilation, wall-clock and comparable classical compute.

401 **Classical validity.** Coordinate greedy and cluster repair are meaningful controls but not optimality
402 certificates for the full 12/18-task problem. The exact certificate applies only to the retained core
403 conditional on the outside incumbent. Future work should add mixed-integer, constraint-programming
404 and stronger heuristic portfolios with equal tuning budgets.

405 **Potential benefits and harms.** Explicit abstention, audit burden and human-mandatory modes
406 could support accountable system design. The same optimizer could also rationalize labor substitution,
407 centralize authority, or disguise a contested value judgment as a technical coefficient. Mitigations
408 include participatory specification, disaggregated outcome reporting, review-capacity stress tests,
409 appeal/override channels, privacy protection and independent impact assessment. Standards such
410 as NIST AI RMF and ISO/IEC 42001/23894 provide governance vocabulary, but compliance and
411 conformity assessment are outside this study.

412 **Anonymous and camera-ready artifacts.** The compiled submission omits an identifying URL to
413 preserve double blindness. Before camera-ready release, authors should insert the permanent public
414 URL and commit, verify its license, archive the exact artifact, and ensure the paper does not imply
415 that governance references endorse the benchmark.

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419 contributions and scope?

420 Answer: [\[Yes\]](#)

421 Justification: The abstract, Introduction, Results, and Table 2 state the conditional statevector claim
422 and explicitly exclude QPU, runtime, scaling, safety, and universal-advantage claims.

423 Guidelines:

- 424 • The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the
425 paper.
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427 made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this
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430 results can be expected to generalize to other settings.
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432 attained by the paper.

433 **2. Limitations**

434 Question: Does the paper discuss the limitations of the work performed by the authors?

435 Answer: [\[Yes\]](#)

436 Justification: Section 6 and Appendix F discuss synthetic construct validity, classical controls, small
437 retained cores, unequal searches, noiseless simulation, and societal risks.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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